

Section I: Cover Page

Title: Cool Med Device Inc. Quality Manual

Document Number: POL-QUAL-00001

Document Version: 1.0

Effective Date: 08-AUG-2024

Reviewers:

Review Function	Name
Executive Level Reviewer	Rod Freeman
Head of Quality	Joseph Quality
Technical	Suva Narayan
Technical	Shelby Reed
Technical	Saadat Ali
Technical	Ben Tran

Version History

Version	Description & Reason for Change
1.0	New document; First version

Section II: Table of Contents

Section I: Cover Page	1
Section II: Table of Contents	2
Section III: Quality Policy Statement.....	3
Section IV: Introduction	3
Section V: Organization	3
Section VI: Facilities	3
Section VII: Quality System	5
Section VIII: Management Responsibility	6
Section IX: Change Control & Change Management.....	8
Section X: Document Control.....	10
Section XI: Vendor Selection, Qualification, and Management.....	12
Section XII: Material and Product Identification and Processing.....	14
Section XIII: Process Control Programs.....	16
Section XIV: Inspection and Testing.....	18
Section XV: Calibration Programs	21
Section XVI: Product Inspection and Test Status.....	23
Section XVII: Control of Non-Conforming Product.....	25
Section XVIII: Corrective and Preventive Action Plans.....	27
Section XIX: Handling and Storage of Materials and Products.....	29
Section XX: Control of Quality Records	31
Section XXI: Internal Quality Audits.....	33
Section XXII: Training Programs	34
Section XXIII: Statistical Techniques	35
Section XXIV: Computer System Controls for GxP Activities	37
Section XXV: Facility Control	39
Section XXVI: Management of Equipment.....	41
Section XXVII: Computer System Validation.....	43
Section XXVIII: Occupational Health and Safety Programs.....	45

Section III: Quality Policy Statement

Cool Med Device Inc. is committed to deliver products and services that meet or exceed requirements and expectations of our customers, stakeholders, and regulatory bodies. Our approach is guided by robust quality management systems, adherence to regulatory requirements, and a relentless focus on customer satisfaction. We are dedicated to maintaining the integrity, safety, and effectiveness of our products, from initial design through to final delivery, while continuously improving our processes to enhance performance and minimize risk. In our own desire to constantly improve and remain above the competition we ensure that quality is embedded in everything we do, driving our commitment to excellence.

Section IV: Introduction

This Quality Manual defines the key elements of the Quality Management System of Cool Med Device Inc. It outlines the policies, processes, and procedures that ensure our products meet the highest standards of quality, safety, and effectiveness. It serves as a comprehensive guide for maintaining compliance with regulatory requirements and applicable international standards.

This Quality Manual is applicable for all aspects of the medical device manufacturing activities/operations. It encompasses the entire lifecycle of Cool Med Inc.'s medical devices from product design and development through production, distribution, and post-market surveillance.

Section V: Organization

There are several departments contributing to overall company workflow at Cool Med Device Inc., ranging from product design, manufacturing, validation, and regulatory affairs. However, there are three key teams that are critical for ensuring the company's success across all departments.

Executive Management - Executive management is responsible for critical decisions that will determine the success and growth of Cool Med Device Inc. Key members are the Chief Executive Officer (CEO), Chief Operating Officer (COO), and members within the Board of Directors.

Quality Management - Quality management reports directly to executive management and is responsible for establishing the robust quality management system that is outlined in this quality manual. Quality management's most crucial priority is ensuring that every element of operations taking place at Cool Med Device Inc. meets and adheres to not only standards and specifications set by the company, but to regulations enforced by federal law (i.e. US Food and Drug Administration, 21 C.F.R.).

Information Technology (IT) - IT personnel are integral components of Cool Med Device Inc. that work alongside ALL departments to ensure that all technology and tools used in company operations are fully functional for optimal workflow support. Additionally, IT will work accordingly

with quality management for the oversight of computer system validation, a critical element in evaluation of GxP compliance across Cool Med Device Inc. operations.

Section VI: Facilities

Cool Med Device Inc., is located in Baltimore, Maryland, and operates in a space that supports the type(s) of work that are performed at this facility. The facility is a 90 acre campus where all Class I, II, and III medical devices are tested. Additionally, our facility offers Physicochemical Evaluations, Microbiology Safety, Stability Studies, and Biological Safety testing for all Class I, II, and III Medical Devices that enter our facility.

The conditions must not impact the results of the work carried out or impact the quality of operations at Cool Med Device Inc. The following controls are monitored at Cool Med Device Inc. to ensure no impact on all work/operations:

- Preparation and Medical Device Testing Labs are not directly open to the outside.
- All windows are not open unless for an emergency.
- Restricted access to all facilities and these areas must not be entered by unauthorized personal or untrained.
- Building security to protect the safety and confidentiality of all employees and visitors as well as the work performed at the facility.
- Free of infestation of insects, rodents, birds, or other vermin.
- Cleaning of all restrooms, break areas, offices, hallways, and conference rooms by in-house personnel.
- Control and monitoring of environmental contamination levels.
- Procedures that describe and outline all clothing and PPE.
- Access to all appropriate hand washing and sanitation stations.
- HEPA filtration to all laboratory spaces.
- Monitoring and control of temperature and humidity.
- All labs and office spaces are equipped with fire protection equipment and a fire alarm notification system.
- Maintenance and monitoring of all water systems for testing operations.

Cool Med Device Inc. is responsible to ensure that the facility is properly equipped with the appropriate equipment/instrumentation along with the appropriate software needed to perform testing activities for the Medical Devices tested at our facility.

All equipment and software that is used in the testing process at Cool Med Device Inc is evaluated and assessed to ensure that the equipment and software can be performed with precision and accuracy for testing purposes. All testing and measuring equipment is referenced on a master equipment list.

All equipment and software that enters Cool Med Device Inc. is assigned a unique identification number for tracking purposes, is routinely maintained, calibrated, or qualified based on work

function, an instrument tag to reflect current status of instrument (i.e., specific operating conditions or removed from service), and is safeguarded.

All documentation to any qualification/calibration/maintenance events or equipment changes must be documented in real time and should capture and reflect the current status.

All personnel will need to be trained in order to operate the appropriate equipment/software as well as performing maintenance, calibration, or qualification for that equipment/software.

Section VII: Quality System

Introduction and Overview:

The quality system at Cool Med Device Inc. is designed to ensure that our products consistently meet regulatory and customer requirements for safety, quality, and performance. This section provides a high-level review of the quality system, including responsibilities, structure, and management's role.

Level 1 Quality Policy:

Our Quality Policy outlines our commitment to delivering products of the highest quality. It serves as the foundation of our quality system, guiding every decision and action within the company. The policy emphasizes customer satisfaction, regulatory compliance, continuous improvement, and the importance of each employee's role in achieving these goals.

Level 2 Quality Manual:

The Quality Manual is the document that defines the overall structure of our quality system. It describes the organization's approach to quality management, including the processes and procedures that ensure our products are consistently manufactured to the highest standards. This manual also outlines the roles and responsibilities within the quality system, ensuring clarity and accountability at all levels.

Level 3 Standard Operating Procedures:

Standard Operating Procedures (SOPs) provide detailed instructions on how to perform specific tasks within the quality system. These procedures ensure consistency, accuracy, and compliance with both regulatory requirements and internal standards. SOPs cover every aspect of our operations, ensuring that all activities are performed correctly and efficiently.

Level 4 Quality Records:

Quality Records are the documented evidence that our quality system is functioning as intended. These records include data and documentation generated during the execution of SOPs, such as inspection results, testing data, and audit reports. Quality Records are maintained and reviewed to ensure traceability, compliance, and continuous improvement.

Responsibilities and Structure:

Quality at Cool Med Device Inc. is everyone's responsibility. The Quality Assurance (QA) department leads the effort, with the Quality Manager serving as the primary authority on quality matters. QA collaborates closely with all departments to implement and maintain the quality system, ensuring compliance with regulatory standards and internal policies.

The quality system is structured to support oversight and continuous improvement. This structure includes dedicated teams responsible for key quality functions such as document control, process validation, and supplier management. Each team is tasked with specific responsibilities that contribute to the overall effectiveness of the quality system.

Management's Role:

Management at Cool Med Device Inc. plays a crucial role in the quality system. They are responsible for setting quality objectives, providing the necessary resources, and fostering a culture of quality throughout the organization. Management is also actively involved in the review and improvement of the quality system, ensuring that it remains effective and aligned with the company's strategic goals.

Regular management reviews are conducted to assess the performance of the quality system. These reviews occur at least annually, but more frequently if needed, and include an evaluation of key quality metrics, audit results, customer feedback, and regulatory updates. The outcomes of these reviews drive continuous improvement initiatives and help ensure that the quality system evolves in response to changing conditions and requirements.

Section VIII: Management Responsibility

Introduction & Overview

The Management team at Cool Med Device Inc., has been actively involved in the implementation of the Quality Management System. The Quality Management System provides the strategic direction and vision of the growth of the Quality Management System, the quality policy, and the quality objectives.

Scope

The section outlines the responsibility and roles of management at Cool Med Device Inc. in the quality process. The Management team is responsible for ensuring the availability of resources that are required in order to maintain and improve quality and operational management systems. The Management team also is responsible for providing leadership that is needed in order to be functionally successful for the Quality Management System.

Applicable SOPs

- SOP-QUAL-0404: Quality Risk Management
- SOP-QUAL-0396: Data Integrity and Management
- SOP-QUAL-2024: Quality Management
- SOP-QUAL-1996: Organization Chart

Policy

- Executive Management strives for the core values of Cool Med Device Inc. These core values include quality, customer focus, integrity, and team spirit.
- Quality: delivering quality across all work performed while also providing accurate results, using appropriate methods and technology, and seeking to change or improve processes for future work.
- Customer Focus: aiding to provide value for our customers through the services provided through Cool Med Device Inc., continuing to innovative solutions to help our customers achieve their goals, and continuing to exceed customer expectations.
- Integrity: Always ensuring Cool Med Device Inc. is operating in a responsible work environment, showing respect to all customers and employees, and behaving ethically at all times.
- Team Spirit: encouraging and recognizing employees for outstanding performance, investing in an employees training and growth for career opportunities, and employing a staff that is competent, and talented.

Roles and Responsibilities

Role	Responsibilities
Executive Management	• Appoints management, quality, and technical personnel appropriately to implement, improve, and maintain the Quality Management System. • Ensure that all personnel are free from any undue financial, commercial, or other pressure that potentially could influence their quality of work or decision making. • Provide a safe environment for all employees. • Ensure good communication is present across all of Cool Med Device Inc. on overall performance and effectiveness of the Quality Management System. • Guarantee the protection of intellectual property and confidential information. • Establish the organizational structure of Cool Med Device Inc. • Make aware to all personnel through promotion and training how their activities impact the growth of the Quality Management System.

Section IX: Change Control & Change Management

CHANGE CONTROL PROCESS

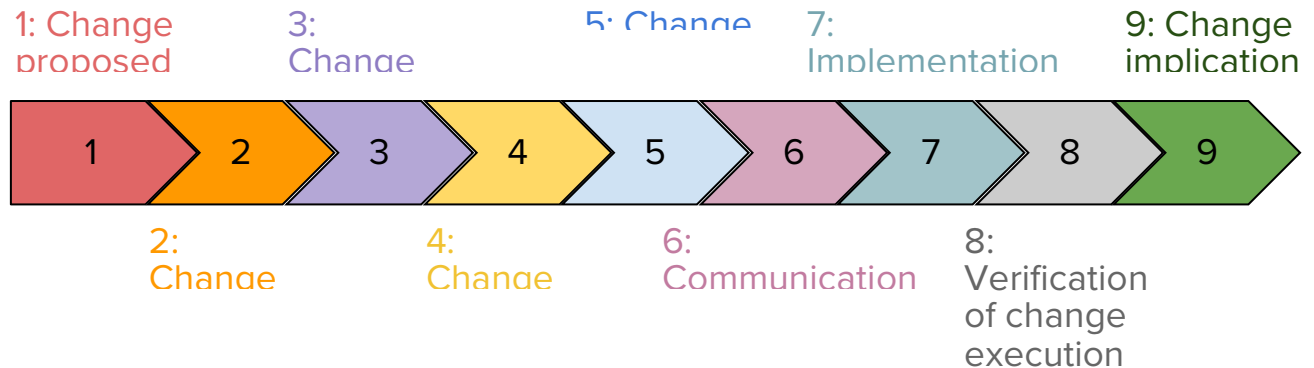


Figure 1: Change control lifecycle. Proposed and anticipated changes to any operation, product, or process at Cool Med Device Inc. must undergo this lifecycle to uphold quality standards and for best enforcement of compliance amongst personnel.

Introduction & Overview

Change control facilitates and streamlines the process of applying changes, regardless of magnitude, to any and all of Cool Med Device Inc.'s operations. It is imperative to Cool Med Device Inc.'s quality management system that robust protocols are in place to ensure effective change control and change management throughout company functions.

Scope

This section details Cool Med Device Inc.'s general practices for implementing change control and change management across all departments and operations. Quality management defines change control as a system for evaluation of modifications and adjustments to a procedure, product, or service.

Applicable SOPs

- SOP-QUAL-0408: Change Control Procedure
- SOP-QUAL-0417: Change Management Protocols
- SOP-QUAL-0418: Validation of Change Control

Policy

- Proposed or intended changes to existing protocols, operations, products, and functions must follow the general timeline for change control to ensure that proposed change is effectively implemented once approved
- Evaluation and assessment of intended changes must be supported by the corresponding department to enhance effective implementation and communication of change upon approval
- It is imperative for company operations that approved changes to any protocol, product, or function are sufficiently and appropriately communicated to all personnel of the corresponding department, and that the change is tracked and monitored to review the long-term implications on the department as well as the overall company
- Quality change management is dependent on the effectiveness of the change control timeline. As long as all intended change requests & proposals are properly reviewed, assessed, and approved if acceptable, the change management system will be centered around the timeline and straightforward.
- For changes to protocol and data documents such as standard operating procedures, ensure that version number is changed and updated as well as the effective date. This will ensure all personnel are up to date and refer to the most recent versions of protocols to follow.

Roles & Responsibilities

Role	Responsibilities
Initiator	● Must come from department requesting change ● Responsible for providing adequate reasoning and evidence to support need for change
Subject Matter Expert	● Works with initiator to provide reasoning and justification for necessity of change ● Works to ensure that change is not proposed out of negativity for current working protocols, and that it is proposed as a legitimate improvement to operations of the department and the company as a whole
Quality Assurance of Change Management	● Conduct preliminary assessments of change to initially screen for potential and risks ● Perform thorough evaluation of change justifications prior to final procedures for approval or rejection of change
Departmental	● Ensure adequate communication of new change across all employees and functional groups within department ● Oversee and streamline implementation of new change, and record observations as change becomes effective ● Work with change management and quality assurance to validate change's execution and evaluate long-term effectiveness and implications of change

Section X: Document Control

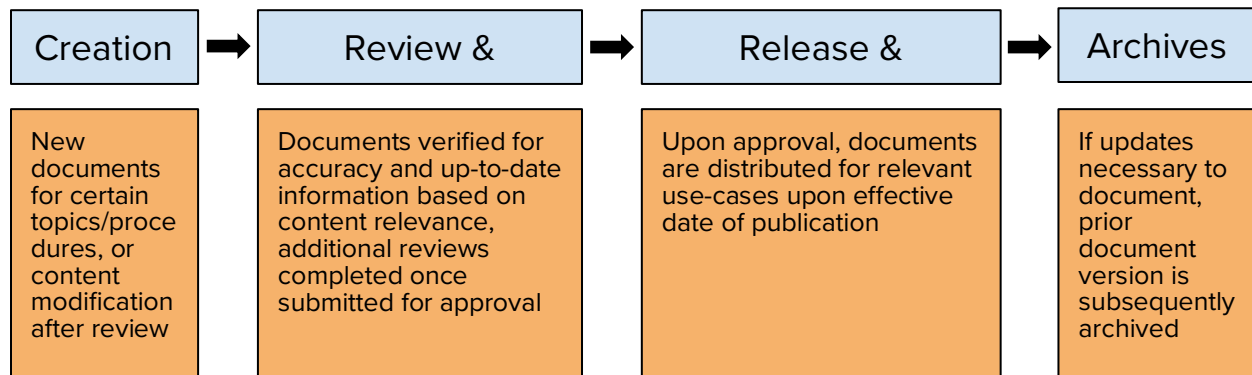


Figure 2: Document control lifecycle. Creation and modification of all documents across departments at Cool Med Device Inc. must go through a cycle of approval, publishing, and archiving. This ensures compliance with up-to-date standards amongst personnel, and provides references for quality management to analyze changes between newer and older documents.

Introduction & Overview

It is important that documents are identified, handled, and defined accurately and carefully for the critical upkeep of data and informational integrity across Cool Med Device Inc.'s employees and divisions.

Scope

This section details Cool Med Device Inc.'s general practices and protocols for document control. Quality management has defined these protocols critical to the overall quality management system, to maintain uniformity and promote data integrity across all departments.

Applicable SOPs

- SOP-QUAL-0419: Document Control Process & Lifecycle
- SOP-QUAL-0408: Change Control Procedure

Policy

- Any form of documentation relevant to operations within Cool Med Device Inc. must undergo the general document control cycle, detailed in Figure 2 above.
- Quality assurance and control must oversee the establishment of an effective system for document control for each department within Cool Med Device Inc, and documents must be stored within the company's document management system. Changes to this system must be reported to quality management.
- ALL documents within the document management system must be uniquely identified, and clearly indicate an effective date as well as a version identifier to indicate the existence of revision history.
- Documents must be reviewed and approved by authorized committees prior to their use within any department.

- To ensure that revision history is tracked, all changes to documents must be tracked via change control timeline (detailed in Section IX), and identified appropriately in the document control cycle for upkeep of revision history.
- Documents with obsolete information (i.e. prior versions, old procedures) must be archived appropriately by department name.

Roles and Responsibilities

Role	Responsibilities
IT	• Responsible for proper implementation and maintenance of document management system • Resolves all calibers of technical difficulties
Quality Assurance	• Completes review and suggests changes to preliminary document versions before submission for approval • Responsible for verification of accuracy and up-to-date content respective to specific department of Cool Med Device Inc.
Quality Control and Process Engineers	• Distributes documentation to respective departments upon approval of document version • Responsible for proper archiving of older versions or obsolete formats of documents • Communicate needs for change to any documentation that requires editing to Quality Assurance

Section XI: Vendor Selection, Qualification, and Management

Introduction & Overview:

This section describes Cool Med Device Inc.'s process for identifying, selecting, qualifying, and lifecycle management of contract service providers and materials suppliers. Quality management and quality operations oversight of contract service providers and material suppliers ensures that controls are in place to maintain product quality and compliance with applicable statutes and regulations.

Scope:

- This policy applies to Cool Med Device Inc.'s management of material suppliers (e.g. raw material manufacturer) and service providers (e.g. contract testing labs) supporting or directly conducting GMP operations for or on behalf of Cool Med Device Inc.
- Material suppliers and service providers for Cool Med Device Inc. manage their own material suppliers and service providers; the oversight of the latter suppliers and service providers is out of scope.

Applicable SOPs:

- SOP-QUAL-0400: Supplier Selection & Justification
- SOP-QUAL-0401: Supplier Qualification
- SOP-QUAL-0402: Supplier Corrective Action Request
- SOP-QUAL-0403: Audits and Self-Inspection
- SOP-QUAL-0404: Quality Risk Management

Policy:

- Cool Med Device Inc. uses a risk- and performance-based approach to the selection and management of material suppliers and service providers.
- Risk is assessed as impact to the finished product's quality and this risk assessment is used to classify material suppliers and service providers into different tiers (see SOP-QUAL-0401: Supplier Qualification).
- The tier classification is then used to determine phase-appropriate requirements for supplier qualification and management. Supplier/provider oversight varies depending on risk to product and phase of product development.
- Material suppliers and service providers are managed through routine performance monitoring, quality audits, and quality and material supply agreements.

Roles & Responsibilities:

Role	Responsibilities
Initiator	• Works with Supply Chain and Subject Matter Expert(s) to identify potential suppliers

	• Initiates requests to perform qualification of material supplier or service provider • Provides justification for the selection of supplier/provider • Leads risk assessment
Subject Matter Expert	• Collaborates with the initiator to evaluate and determine the technical requirements for the material/service • Reviews the technical aspects of quality agreements • Participates in audits as needed
Quality Assurance	• Approve or reject selection and qualification decisions • Review and approve supplier risk mitigation plans
Supplier Quality Management	• Oversee supplier quality management program • Evaluate and approve suppliers for their ability to perform work in accordance with regulatory and quality requirements • Conduct audits of suppliers/providers • Proposes supplier/provider selection and qualification decisions • Establish and review quality agreements • Monitor performance and maintain quality oversight of suppliers • Develop risk mitigation plans • Report out to management quality issues
Supply Chain	• Works with the Initiator and Subject Matter Expert(s) to identify potential suppliers • Review and approve material supply agreements • Manage material inventory based on material risk assessments

Section XII: Material and Product Identification and Processing

Introduction & Overview

Traceability is a crucial aspect to ensure product safety and compliance with regulatory requirements for every medical device that is tested and developed at Cool Med Device Inc. This process involves record keeping, adherence to current regulatory standards, and advanced technology.

Scope

The purpose of this section is to outline the material and product identification and processing system at Cool Med Device Inc. These material and product identification and processing systems help to keep traceability of all records within Cool Med Device Inc. for compliance and quality purposes.

Policy

- Unique Device Identification (UDI): all medical devices entered at Cool Med Device Inc., will be assigned a UDI that will include a device identifier (DI) and production identifier (PI). The DI will be specific to the device model and the PI will list lot information, serial number, and expiration date.
- All medical devices will be labeled with their UDI on both the packaging of the device as well as the device itself.
- All medical devices will be labeled with a barcode and a radio-frequency identification tag (RFID) that will display all information about the medical device and advanced tracking for inventory management and internal tracking at Cool Med Device Inc.
- Documentation and recordkeeping of all medical devices will be stored in a Device Master Record (DMR) and a Device History Record (DHR)
- The DMR contains information that is for the production of the medical device. This information includes packaging, product specifications, procedures for quality assurances, and processes for manufacturing.
- The DHR contains information for each batch/unit of medical device that comes through Cool Med Device Inc. This information includes certificate of analysis, production/product history, quantity manufactured, and manufacturing dates.
- All suppliers and vendors approved at Cool Med Device Inc will have regular audits to ensure that they are within compliance and quality standards for all products used at Cool Med Device Inc. in order to ensure traceability.
- The Laboratory Information Management System (LIMS) will be used to report and store all Certificate of Analysis (CofA), CAPAs, Non-routine Events (NREs), Out of Specifications (OOS), Planned/Unplanned Deviations for all reagents, medical devices, vendors, instruments, and consumables that come through Cool Medical Device Inc.
- All data that is considered to be critical will be stored in a secure area that allows for routine backup, retrieval, and archival.

Applicable SOPs

- SOP-QUAL-0403: Audits and Self-Inspection
- SOP-QUAL-1989: Document Control
- SOP-QUAL-0400: Supplier Selection & Justification
- SOP-QUAL-0401: Supplier Qualification
- SOP-QUAL-0408: Change Control Procedure
- SOP-QUAL-0417: Change Management Protocols
- SOP-QUAL-0418: Validation of Change Control
- SOP-QUAL-2024 Quality Management System

Roles and Responsibilities

Role	Responsibilities
Quality Assurance	• Ensures that any audit, investigation, OOS, NRE, CAPA, and deviations are referenced to medical devices, suppliers/vendors, reagents, or equipment at Cool Med Device Inc. • Ensures traceability for all samples that enter Cool Med Device Inc. • Ensures traceability of all sample data and CofA's generated at Cool Med Device Inc. prior to being released to the client.
Quality Control	• Reports all nonconformities as needed to quality assurance • Ensures all training records are up to date. • Measures and monitors results that are applicable in the production process.
Quality Control Document Control	• Ensures all documents are archived appropriately. • Performs routine back-ups, archival, and retrieval for all data generated at Cool Med Device Inc.

Section XIII: Process Control Programs

Introduction & Overview:

Process control programs ensure that all manufacturing processes at Cool Med Device Inc. are consistently producing products that meet predetermined specifications for quality, efficacy, and safety. These programs encompass the procedures for validating, monitoring, and controlling the manufacturing processes to ensure conformity with regulatory requirements and internal standards.

Scope:

This policy applies to all manufacturing processes at Cool Med Device Inc, including process validation, monitoring, and control activities. The policy for managing changes to processes and handling nonconformities is also included.

Applicable SOPs:

- SOP-QUAL-1997: Process Validation
- SOP-QUAL-1998: Process Monitoring and Control
- SOP-QUAL-0440: Corrective and Preventive Actions (CAPA)
- SOP-QUAL-0441: Nonconforming Product Control
- SOP-QUAL-0408: Change Control Procedure
- SOP-QUAL-XXXX: Equipment Calibration and Maintenance

Policy

- Process Validation: All new processes, as well as any changes to existing processes, must be validated before implementation. Validation ensures that processes consistently produce outputs that meet specified criteria.
- Process Monitoring: Continuous monitoring of processes is conducted to detect any deviations from specified parameters. Monitoring includes periodic testing and inspection at critical control points.
- Process Control: Defined procedures are in place to control the various aspects of the manufacturing process. These controls include equipment calibration, maintenance, and regular review of process performance data.
- Nonconforming Product Management: Any products that do not meet quality specifications are identified, segregated, and investigated. Appropriate corrective actions are taken to address the root causes of nonconformities.
- Change Control: All changes to processes, equipment, or materials are managed through a formal change control procedure to ensure that the changes do not adversely affect product quality.
- Documentation: Detailed records of process validation, monitoring, and control activities are maintained to demonstrate compliance with regulatory requirements and to facilitate continuous improvement.

Roles and Responsibilities

Role	Responsibilities
Quality Assurance	• Oversees the implementation and maintenance of process control programs. • Reviews and approves process validation protocols and reports. • Conducts regular audits of process control activities to ensure compliance.
Process Engineer	• Develops and documents process control plans. • Performs process validation and re-validation as necessary. • Analyzes process performance data to identify areas for improvement. • Implements corrective and preventive actions to address process deviations.
Production Manager	• Ensures that production processes are conducted in accordance with established process control plans. • Trains production staff on process control procedures and best practices. • Monitors daily production activities to ensure adherence to process specifications. • Reports any process deviations to QA for investigation and resolution.
Quality Control Technician	• Conducts routine inspections and tests to verify process outputs. • Records and reports test results accurately and promptly. • Identifies and documents nonconforming products and processes. • Supports process validation and monitoring activities as required.
Regulatory Affairs Specialist	• Ensures that all process control activities comply with relevant regulatory requirements. • Maintains up-to-date knowledge of regulatory changes and communicates them to the team. • Supports the preparation of regulatory submissions related to process changes and validations.

Section XIV: Inspection and Testing

Introduction & Overview

At various stages of production, it is important that quality is ensured at every stage of the raw material receipt, in-process activities, and prior to shipment. However, there are at times during these stages that a process would need to be rejected. This is crucial in our business as we strive to maintain high standards of quality and customer satisfaction.

Scope

The purpose of this section is to outline the inspection and testing procedures at Cool Med Device Inc. in order to ensure quality. This section will also outline how Cool Med Device Inc. handles process rejection.

Applicable SOPs

- SOP-QUAL-0403: Audits and Self-Inspection
- SOP-QUAL-0404: Quality Risk Management
- SOP-QUAL-0405: Product Lot Disposition
- SOP-QUAL-0406: Relabeled Product Disposition
- SOP-QUAL-0407: Deviation Procedure
- SOP-QUAL-0407: Change Control Procedure
- SOP-QUAL-0408: Rework and Reprocessing
- SOP-QUAL-0409: Handling, Removal, Disposal Procedure
- SOP-QUAL-1997: Process Validation
- SOP-QUAL-1998: Process Monitoring and Control
- SOP-QUAL-0440: Corrective and Preventive Actions (CAPA)
- SOP-QUAL-0441: Nonconforming Product Control

Policy

- When raw materials are received at Cool Med Device Inc. all raw materials will be subjected to the following:
 - Supplier Certification: All suppliers for raw materials will be certified to show they have a history of providing Cool Med Device Inc. quality raw materials
 - Visual Inspection: All raw materials will be subjected to visual inspections to check for damages or defects. The visual inspection will also include a dimensional inspection to ensure the raw material meets all dimensions and tolerances listed in the raw materials paperwork submitted by the supplier.
 - Raw Materials Testing: All raw materials will undergo both physical and chemical testing utilizing cGMP and GLP standards to ensure that the raw material meets the requirements for its physical and chemical compositions.
 - Documentation Review:
- All raw materials will be subjected to In-Process Activities at Cool Med Device Inc. The following In-Process Activities are:

- Process control plan: The process control plan will outline all critical control points and inspection criteria.
- First Article and In-Process Inspections: These inspections will occur with the first raw material produced as well as during the manufacturing process.
- Statistical Process Control: Statistical methods will be utilized to monitor and control processes.
- Operator Check-Ins: Operators will report any deviations and perform self-checks.
- Non-Conformance Management: all non-conformities will be identified, documented, and addressed within Cool Med Device Inc's LIMS system.
- Prior to shipment to Cool Med Device Inc. the client will be responsible for the following procedures for its raw materials:
 - Final Inspections: A visual final inspection of the raw material will be completed to check for defects.
 - Quality Documentation: Client will complete a raw material submission form and will ensure that the form is complete and accurate.
 - Packing Inspection: All packaging will be inspected to ensure adequacy to ensure the raw material is protected during transport to Cool Med Device Inc.
- If a process is subjected to rejection, the process will go through the following rejection process:
 - Identification: Identify the product or process that did not meet quality standards or specifications.
 - Documentation: Ensure that all non-conformances are documented this includes the impacted batch/lot, defects, and test results.
 - Corrective Action: Develop and implement a corrective action that addresses the root cause and prevents the non-conformity from recurring.
 - Re-evaluation: Once corrective actions have been implemented, re-evaluate the process and ensure that all quality standards are now met.
 - Communication: Communicate the corrective actions and rejections to all production, management, quality assurance, and stakeholders.

Roles and Responsibilities

Role	Responsibilities
Quality Assurance	● Keeps records of all suppliers and vendors. ● Reviews all inspections, investigations, deviations and CAPAs due to issues related to testing and identification.
Quality Control	● Performs all testing with quality documentation and proper training. ● Reports any non-conformities to executive management.
Overchecker	● Double-checks all work performed by Quality Control prior to

	CofA release.
Executive Management	• Reports any non-conformities to Quality Assurance. • Works with Quality Assurance on effective CAPAS.
Client	• Ensures that all medical devices are submitted with the correct paperwork to Cool Med Device Inc.

Section XV: Calibration Programs

Introduction & Overview

Cool Med Device Inc.'s manufacturing operations and facilities are heavily reliant on the proper calibration of required equipment and instruments used throughout the processes taking place. Quality management defines calibration as protocols to ensure accurate and precise measurements of data integral to specific processes, such as manufacturing and packaging.

Scope

This section provides a detailed description and understanding of the calibration programs and protocols for all relevant instruments used in manufacturing operations for Cool Med Device Inc. Calibration is crucial to quality control, ensuring that the finished product meets specifications detailed by Cool Med Device Inc.'s product development team.

Applicable SOPs

- SOP-QUAL-0420: Equipment Calibration Procedures
- SOP-QUAL-0421: Documentation Guidelines for Equipment Calibration
- SOP-QUAL-0422: Handling of Products Manufactured with Uncalibrated Equipment

Policy

- Instruments and equipment used in the manufacturing operations of all Cool Med Device Inc. products must be routinely calibrated for optimum performance, ensuring that finished products meet specifications in place by the product's corresponding design team.
- Frequency of calibration will depend on the manufacturing facility, but must be specified by the manual for the instrument/equipment, the product design team that will employ the instrument/equipment, as well as the corresponding standard operating procedure (Equipment Calibration Procedures).
- Accuracy and precision limits will be dependent on the specific instrument and equipment.
 - For example, if a certain device is responsible for recording product dimensions, the calibration must ensure that the product meets the measurements specified by the product design team at >90% accuracy.
- If instruments and equipment are not properly and routinely calibrated throughout manufacturing operations, all personnel must follow guidelines in the standard operating procedure for handling of products manufactured with such equipment (Handling of Products Manufactured with Uncalibrated Equipment).
 - Quality control and quality management will oversee this process to ensure that Cool Med Device Inc.'s quality standards are maintained throughout product life cycles.

Roles & Responsibilities

Roles	Responsibilities
Product Design Teams	• Responsible for clearly indicating and defining product specifications and characteristics prior to the start of the manufacturing. • Work with manufacturing personnel to ensure that equipment is calibrated for desired accuracy and precision for optimal product outcomes.
Manufacturing Personnel	• Responsible for calibration protocols. Ensuring that all equipment and instruments are adequately and properly calibrated prior to the manufacturing process of all products. • Works with product design teams and quality assurance to verify and validate manufactured products to desired specifications • Adjust calibration of instruments and equipment as necessary, should the manufacturing process be modified
Quality Assurance & Management	• Works with product design teams and manufacturing personnel to verify and validate manufactured products. • Responsible for facilitating communication between product design teams and manufacturing personnel of product specifications ○ This ensures that manufacturing personnel can prepare facilities for production through proper calibration of employed instruments and equipment for consistency of product quality

Section XVI: Product Inspection and Test Status

Introduction & Overview:

Product inspection and testing procedures ensure that product manufactured at Cool Med Device Inc. conforms to the specification criteria for identity, strength, quality, and purity and that they are properly labeled and packaged.

Scope:

- This policy applies to the inspection, testing, and disposition of products at Cool Med Device Inc.
- The policy for inspection, testing, and disposition of raw materials can be found in Section XIV.

Applicable SOPs:

- SOP-QUAL-0405: Product Lot Disposition
- SOP-QUAL-0406: Relabeled Product Disposition
- SOP-QUAL-0407: Deviation Procedure
- SOP-QUAL-0408: Change Control Procedure
- SOP-QUAL-0409: Rework and Reprocessing
- SOP-QUAL-0410: Handling, Removal, Disposal Procedure
- SOP-QUAL-0411: Laboratory Investigation Procedure
- FORM-QUAL-0401: Product Certificate of Conformance
- FORM-QUAL-0402: Lot Disposition Form

Policy:

- Product inspection and testing controls should be comprehensive of applicable points throughout the manufacturing process, including incoming material, in-process, and release testing.
- Final product testing can include (but is not limited to): physical, chemical, microbiological, functional/performance, and stability testing.
- Specification criteria for product testing and inspection and product label specifications are documented and assessed for compliance with regulatory filings.
- Changes to specifications must follow change control procedures (SOP-QUAL-0408).
- Material labeling and segregation controls are in place to ensure disposition status of product is clear in the manufacturing facility. The following statuses are used:
 - *Quarantine*: Materials or products awaiting inspection or test results. Quarantine materials are kept in a designated area and are not released until all evaluations are complete, results meet the defined acceptance criteria, and are dispositioned by Quality Assurance.
 - *Released*: Products that have passed all required tests and inspections are labeled as released and are dispositioned by Quality Assurance.

- *Held*: Products that have been temporarily removed from the production or distribution process due to a suspected non-conformance event or a pending investigation (e.g. as part of a deviation).
- *Rejected*: Products that have failed to meet quality specifications and are deemed unsuitable for use. These products are appropriately labeled and disposed of (see SOP-QUAL-0410: Handling, Removal, Disposal Procedure) or reprocessed (see SOP-QUAL-0409: Rework and Reprocessing).
- Products that are out of specification must be investigated per established procedures (SOP-QUAL-0411: Laboratory Investigation Procedure and SOP-QUAL-0407: Deviation Procedure).

Roles & Responsibilities:

Role	Responsibilities
Manufacturing	● Review product batch record documents and device history files
Quality Assurance	● Review and evaluate product batch record documents and device history files ● Determines and documents the product disposition decision and attaches documentation to the batch production record
Quality Control	● Performs testing as specified for each product ● Generates, reviews, and approves associated product-specific test result documentation ● Generates, reviews, and approves a certificate of testing
Quality Assurance Document Control	● Archives batch record documents and device history files after disposition

Section XVII: Control of Non-Conforming Product

Introduction & Overview:

Non-conforming products can be identified during any stage of production through routine inspection and testing points in the process. Once identified, non-conforming products are documented, labeled, segregated, and reviewed by Quality Assurance to determine any necessary further action.

Scope:

- This policy applies to the management of non-conforming products at Cool Med Device Inc.
- The policy for inspection, testing, and disposition of raw materials can be found in Section XIV.

Applicable SOPs:

- SOP-QUAL-0405: Product Lot Disposition
- SOP-QUAL-0406: Relabeled Product Disposition
- SOP-QUAL-0407: Deviation Procedure
- SOP-QUAL-0408: Change Control Procedure
- SOP-QUAL-0409: Rework and Reprocessing
- SOP-QUAL-0410: Handling, Removal, Disposal Procedure
- SOP-QUAL-0411: Laboratory Investigation Procedure
- FORM-QUAL-0401: Product Certificate of Conformance
- FORM-QUAL-0402: Lot Disposition Form

Policy:

- Non-conforming products are immediately segregated from conforming products to prevent any mix-up. There must be a dedicated area for non-conforming products
- Each non-conforming product is tagged with a label indicating its status and the reason for non-conformance.
- A cross-functional team reviews the non-conformance report and identifies and documents the proposed action(s), which can include rework, reprocessing, or rejection. Quality assurance must review and approve the actions prior to execution.
- Rejected products are disposed of in a manner that complies with established procedures (SOP-QUAL-0410: Handling, Removal, Disposal Procedure). These procedures must conform with regulatory and environmental guidelines. Disposal is documented, and records are maintained.

Roles & Responsibilities:

Role	Responsibilities
------	------------------

Manufacturing	• Identifies non-conforming products during in-process inspection/testing points
Quality Assurance	• Review and evaluate non-conforming events and product batch record documents and device history files • Approves any proposed actions to rework or reprocess any non-conforming product • Determines and documents the product disposition decision and attaches documentation to the batch production record
Quality Control	• Performs testing as specified for each product • Generates, reviews, and approves associated product-specific test result documentation • Generates, reviews, and approves a certificate of testing • Identifies non-conformant test/inspection results
Quality Assurance Document Control	• Archives non-conformance event document as part of batch record documents and device history files
Materials Management	• Performs disposal of non-conforming product as appropriate and indicated by Quality-approved procedures

Section XVIII: Corrective and Preventive Action Plans

Introduction and Overview

Corrective and Preventive Actions (CAPAs) are used in order to identify and eliminate a non-conformity immediately. By taking a proactive approach and providing a positive work environment Cool Med Device Inc., we are also able to identify preventative actions before they turn into a non-compliant issue.

Scope

The purpose of this section is to outline Cool Med Device Inc.'s CAPA policy and how Cool Med Device Inc. addresses systemic and isolated events.

Applicable SOPs

- SOP-QUAL-0438: Process Validation
- SOP-QUAL-0439: Process Monitoring and Control
- SOP-QUAL-0440: Corrective and Preventive Actions (CAPA)
- SOP-QUAL-0441: Nonconforming Product Control
- SOP-QUAL-0408: Change Control Procedure

Policy

- Executive Management will be responsible for proposing CAPAs while Quality Assurance will be responsible for reviewing CAPAs and ensuring they are implemented in a timely manner.
- Corrective actions will be implemented in order to reduce non-conformities within Cool Med Device Inc.'s technical operations, deviation from our systems, or within our quality system.
 - This will be completed through an investigation to determine the root cause(s) based on the deviation, departure, and non-compliance issue that has occurred.
- Any and all procedures will need to be available to address any corrective action systems/directives
- If more than one area is involved, then collaboration and agreement between Quality Assurance and Executive Management to determine the type of action(s) needed to be implemented for the CAPA.
- If the issue(s) appears to be isolated, it is important that the CAPA will have an impact on the issue and the outcome of the action(s) will be monitored to ensure the CAPA is an acceptable and effective resolution.
- If the issue(s) appears to be repetitive then the affected issue(s)/area(s) will need to be further investigated by Quality Assurance.

Roles and Responsibilities

Role	Responsibilities
Quality Assurance	• Reviews CAPAs written by Executive Management once the root cause(s) of the non-conformity are identified. • Ensures that all CAPAs are implemented in a timely manner. • Ensures CAPA will effectively prevent the non-conformity from happening again. • Monitors all CAPAs at Cool Med Device Inc.
Executive Management	• Reviews data from hypothesis or investigative test plans and deviations to rule out a root cause(s) and proposes a CAPA based on this data.

Section XIX: Handling and Storage of Materials and Products

Introduction & Overview:

Proper product handling and storage of materials and products ensures maintenance of product quality, prevention of contamination or product damage over the shelf-life of the materials and products.

Scope:

- This policy applies to the handling and storage of materials and products manufactured at Cool Med Device Inc.
- The policy for management of non-conforming materials and products can be found in Sections XIV and XVII.

Applicable SOPs:

- SOP-QUAL-0405: Product Lot Disposition
- SOP-QUAL-0406: Relabeled Product Disposition
- SOP-QUAL-0412: Raw Material Lot Disposition
- SOP-QUAL-0407: Deviation Procedure
- SOP-QUAL-0410: Handling, Removal, Disposal Procedure
- SOP-QUAL-0413: Environmental Monitoring Program

Policy:

- Written procedures are established to describe the receipt, identification, quarantining, storage, handling, sampling, testing, and approval or rejection of materials and products.
- All persons involved in material and product handling must be trained on applicable procedures and safety controls.
- Materials and products in storage areas are appropriately labeled with their disposition status (see sections XIV and XVI) and adequately segregated to mitigate the risk of mix-up and cross-contamination.
- Material storage areas must be clean, secure, and access-controlled to prevent unauthorized entry and tampering.
- Only materials and products meeting established specifications and within their expiry are released by Quality can be used/further processed in manufacturing.
- Labeling controls are in place to ensure traceability of materials throughout the facility and lifecycle of a product.
- Materials and products must be stored in controlled and monitored environmental conditions specified in their specifications. Deviations from these storage conditions are investigated for their impact.
- Materials are consumed on a first-in first-out basis (i.e. the oldest materials are used first).

- Periodic inspections of stored products must be conducted to check for degradation and damage. Products nearing their expiration date should be re-evaluated per established procedures
- Comprehensive documentation is maintained for every lot of material.

Roles & Responsibilities:

Role	Responsibilities
Manufacturing	• Handles and stores work-in-progress materials and products as specified in manufacturing work instructions
Quality Assurance	• Determines and documents the raw material and product disposition decision and attaches documentation to the batch production record • Performs routine inspection of the warehouse and material handling and storage areas
Quality Control	• Performs testing as specified for each raw material and product
Quality Assurance Document Control	• Archives documentation for each material/product lot
Materials Management	• Performs the physical receipt and initial inspection of all materials entering the manufacturing facility • Maintains and executes the environmental monitoring program for the warehouse and storage facilities
Facilities & Engineering	• Validates and maintains the HVAC and air handling systems to be within specifications for warehouse and storage facilities for materials and product

Section XX: Control of Quality Records

Introduction & Overview

All quality and technical records must be properly retained such that records can be indexed, collected, identified, securely stored, and accessed conveniently via any department within Cool Med Device Inc. This is to ensure that the integrity of all data within all departments is consistent and maintained, as described in Section X, Document Control.

Scope

This section details Cool Med Device Inc.'s policies and protocols for control, maintenance, and storage of quality records. Quality records, as defined by quality management, are records documenting and constituting the quality of products, operations, services created by operations at Cool Med Device Inc.

Applicable SOPs

- SOP-QUAL-0408: Change Control Procedure
- SOP-QUAL-0419: Document Control Process & Lifecycle
- SOP-QUAL-0423: Quality Record Classification & Maintenance

Policy

- Any and all documentation that contains information regarding the manufacturing process, specifications, and use cases of products developed by Cool Med Device Inc. will be classified as quality records.
- Quality records must follow document control and maintenance guidelines, as outlined in Section X, Document Control. This ensures data integrity and accuracy across the entirety of the company's operations.
- As described in the change control procedure, quality records must be labeled and updated with current version numbers and most recent effective dates.
- As per document control, obsolete versions of quality records must be appropriately archived and organized within that archive by department name, filed by sequential version numbers.
- Quality records must be maintained via a document control program, and all archived records must be kept for a minimum of 10 years for reference.
- Training records constitute training logs, assessments, evaluations, and certificates of relevant personnel within each department.
 - Additional training record management software should be implemented within the document control management system, and follow storage guidelines as outlined earlier.
- Quality record management system must be accessible and easily identifiable by department and version number for quick reference, for quality upkeep and control purposes. This will ensure that documents are consistently up-to-date.

- Should a record require edits or changes (i.e. change in manufacturing or operational procedure), the document must go through the change control team and relevant procedures.

Roles & Responsibilities

Role	Responsibilities
IT	● Responsible for establishment of a robust quality record/document management program for storage of ALL Cool Med Device's records, especially for quality records. ● Ensures that relevant and important employees have access to the system itself as well as the records within the system pertaining to their corresponding department. ● Help personnel should they run into issues working with the management system.
Quality Assurance & Management	● Responsible for enforcement of compliance with quality control. Includes adherence to document and change control process for quality records and ensuring that personnel follow these processes for upkeep of data quality & integrity ● Enforces document management guidelines of record identification and archival if necessary, for record retention
Department and Project Leaders	● Works with quality assurance management to prioritize compliance and regulatory adherence within their respective department amongst all working personnel ● Responsible for creation and maintenance of training records ○ Also ensure that training records are stored within document management system, classified as quality records for reference

Section XXI: Internal Quality Audits

Introduction & Overview:

Internal quality audits are performed regularly to assess compliance with the elements of the Cool Med Device Inc's Quality Management System; regulatory requirements of Good Manufacturing Practices, Good Laboratory Practices, and vigilance practices; and information included in regulatory filings (e.g. INDs, market authorizations).

Scope:

- This policy applies to internal audits performed at Cool Med Device Inc.
- The policy for external audits in the management of contract service procedures and materials suppliers is found in Section XI.

Applicable SOPs:

- SOP-QUAL-0403: Audit and Self-Inspection

Policy:

- Internal audits will be performed by the Quality organization on a regular and defined basis to evaluate and demonstrate compliance Quality Management System; regulatory requirements of Good Manufacturing Practices, Good Laboratory Practices, and vigilance practices; and information included in regulatory filings.
- Auditors must comply with GMP procedures for gowning, access, and environmental health and safety if they enter controlled spaces.
- Audit activities cannot adversely affect routine GMP operations, compromise GMP conditions, or safety of Cool Med Device Inc. personnel.
- In a timely manner, the outcomes of the audits are documented in audit reports, which will include observations made during the audit, proposals for correct actions. Follow-up verification audits are performed as required to ensure corrective actions are adequately implemented.

Roles & Responsibilities:

Role	Responsibilities
Management	• Ensures routine audits are supported and that applicable findings are appropriately addressed in a timely manner
Quality	• Ensures routine audits are performed on a regular and defined basis • Write and issue audit reports
Subject Matter Expert/Key Personnel	• Participates in the audit as an auditee as required

Section XXII: Training Programs

Introduction & Overview

The training program at Cool Med Device Inc. is the basic foundation for all new employees prior to entering their designated departments. This training program is designed to teach new employees the basics of technical and operational skills in their everyday job function.

Scope

The purpose of this section is to outline the training program at Cool Med Device Inc., the frequency of the training, how the training is tracked and stored, and the purpose of the training program.

Applicable SOPs

- SOP-QUAL-1992: Training Program

Policy

- All training requirements can be met through prior job experience, education, on-the job training, specific site training, or internal/external training.
- All training plans/programs need to match the personnel's skill with their job function.
- Training administration will administer all operational and technical training as well as record and maintain all training records for all current employees at Cool Med Device Inc using an electronic document control system.
- Management will assist in the professional growth and development of the employee so that they can maintain their current skill set to their job function and grow in future roles.
- All personnel will receive additional training from their department outside of the basic technical and operational training.

Roles and Responsibilities

Role	Responsibilities
Training Administration	• Administer operational and technical training. • Track and store all training records.
Executive Management	• Review and approve training documentation. • Aid in the planning of an employee's professional growth and development skills.

Section XXIII: Statistical Techniques

Introduction & Overview:

Statistical techniques may be used in the analysis of manufacturing data (e.g. as part of process monitoring), Quality Control data (e.g. QC assay performance monitoring), stability studies, and process characterization and validation studies, among other GMP applications. This policy describes the framework for the application of statistical approaches.

Scope:

- This policy applies to the application of statistical approaches to analyzing GMP data (e.g. as part of GMP process monitoring, QC assay performance monitoring, stability data) and the use of GMP data in process characterization and validation studies and regulatory submissions.
- Out of scope is the prescription for specific statistical approaches as the application of specific statistical approaches may vary depending on the intended use.

Applicable SOPs:

- SOP-QUAL-0414: Process Monitoring Program
- SOP-QUAL-0415: QC Assay Performance Monitoring Program
- SOP-QUAL-0416: Process Characterization and Validation

Policy:

- Statistical approaches in GMP data analysis and process characterization and validation studies must be developed in combination with the appropriate subject matter experts (e.g. statistician).
- Statistical approaches used in quality-approved study protocols, study reports, and SOPs must be explained clearly and justified in said documents so that the analyses can be replicated by a subject matter expert.
- Statistical approaches must be fit for purpose and be based on generally accepted scientific/mathematical principles. Any shortcomings to statistical approaches used (e.g. biases, low resolution) should be discussed in the quality-approved documents.
- Data used for statistical analyses for GMP activities must be verified against source data by a reviewer.

Roles & Responsibilities:

Role	Responsibilities
Quality	• Evaluates statistical approaches and rationale as part of review and approval of the quality-approved study documentation • Determines if a Statistician review and approval is required.

Document Author	• Collaborates with Statistician to determine the appropriate statistical approach for the study and aligns with the Quality reviewer • Authors the study protocol, reports, SOPs
Data Analyzer, SOP/Study Executioner	• Executes the study/SOP • Performs and documents the data analyses using good documentation practices
Statistician	• Collaborates with the document author to define the statistical approach • Drafts the statistical approach procedure, rationale, and discussions as needed • Reviews and approves the technical document as required by Quality

Section XXIV: Computer System Controls for GxP Activities

Introduction & Overview:

The control of computer systems is crucial for ensuring the integrity and reliability of processes, data, and records associated with GxP (Good Practice) activities. This section outlines how Cool Med Device Inc. controls computer systems to ensure accurate process control, data assessment, and compliance with industry regulations. Additionally, it addresses measures for maintaining system security.

Scope:

This policy applies to all computer systems used in GxP activities at Cool Med Device Inc, including systems for manufacturing, quality control, data management, and regulatory compliance. It encompasses system validation, data integrity, access control, and security measures.

Applicable SOPs:

- SOP-QUAL-XXXX: Computer System Validation
- SOP-QUAL-XXXX: Data Integrity and Management
- SOP-QUAL-XXXX: Access Control Procedure
- SOP-QUAL-XXXX: System Security and Maintenance
- SOP-QUAL-XXXX: Backup and Recovery
- SOP-QUAL-XXXX: Change Control Procedure

Policy:

- **Computer System Validation:** All computer systems impacting GxP activities must be validated to demonstrate that they operate as intended and meet predefined criteria. Validation includes installation qualification (IQ), operational qualification (OQ), and performance qualification (PQ).
- **Data Integrity:** Systems must be designed and maintained to ensure the accuracy, consistency, and reliability of data. This includes implementing measures to prevent unauthorized data modification and ensuring traceability through audit trails.
- **Access Control:** Access to computer systems is restricted based on roles and responsibilities. User access levels are defined, and access is granted only to authorized personnel. Regular reviews of user access are conducted to ensure ongoing appropriateness.
- **System Security:** Comprehensive security measures are in place to protect computer systems from unauthorized access, cyber threats, and data breaches. This includes the use of firewalls, antivirus software, encryption, and secure authentication methods.
- **Backup and Recovery:** Regular backups of critical data and systems are performed to ensure data availability in the event of a system failure or data loss. Backup procedures are documented, and periodic recovery tests are conducted.

- Change Control: Changes to computer systems, including software updates, configuration changes, and hardware modifications, are managed through a formal change control procedure to ensure they do not adversely affect system performance or compliance.
- System Maintenance: Routine maintenance activities are scheduled and performed to ensure the continued reliable operation of computer systems. Maintenance records are documented and reviewed.

Roles and Responsibilities:

Role	Responsibilities
IT	● Oversees the implementation and maintenance of computer systems impacting GxP activities. ● Ensures system validation is conducted and documented. ● Manages system security measures and responds to security incidents.
Quality Assurance	● Reviews and approves computer system validation protocols and reports. ● Conducts audits of computer system controls to ensure compliance with GxP requirements. ● Monitors data integrity practices and access control procedures.
System Administrator	● Implements and manages access controls, including user account setup and deactivation. ● Performs regular system backups and recovery tests. ● Maintains system security through updates, patches, and monitoring.
Quality Control	● Utilizes computer systems for data entry, analysis, and reporting. ● Ensures data accuracy and integrity by following established procedures. ● Reports any data integrity issues or system malfunctions to the QA.

Section XXV: Facility Control

Introduction & Overview

Apart from an expansive quality management system, facility management is another pillar that is crucial to governing operations at Cool Med Device Inc. Ensuring and maintaining a safe and secure working environment can contribute positively to workflow and productivity of all personnel.

Scope

This section reviews Cool Med Device Inc.'s standards for facility management across all departments of the company. Quality management defines facility control as the oversight of a facility's infrastructure, operations, and functionalities. Facility control is an important aspect of quality control to ensure that Cool Med Device Inc. can continue to manufacture and deliver products that satisfy manufacturing standards.

Facility management protocols must align with standards specified by ISO 41001, the standard that establishes the requirement of a robust facility management system for all operations of a company.

Applicable SOPs

- SOP-QUAL-0413: Environmental Monitoring Program
- SOP-QUAL-0424: Personal Protective Equipment (PPE)
- SOP-QUAL-0425: Security Protocol
- SOP-QUAL-0426: Sanitation and Sterilization Protocol

Policy

- Facility control standards must be implemented across all departmental buildings of Cool Med Device Inc.
 - Ensure that laboratory workspaces are secure and not freely open to outside areas
 - For buildings with security or restricted clearance, strong protocols must ensure that unauthorized personnel cannot enter the premises
 - Temperatures are optimal and controlled.
 - As appropriate, sanitation guidelines for facility must be in place to ensure sterility
 - Air supply and filtration instruments must be installed and maintained for operational buildings and facilities
 - Rodent and pest infestations must be avoided, procedures to be in place for routine building maintenance
- Building security protocols must ensure the safety and security of all personnel working within facilities across Cool Med Device Inc.

- Protocols must address necessary security clearance and restricted access plans, as well as documentation protocols for visitors, contractors, and service employees (i.e. IT, electricians, equipment repair & calibration) should they need access to restricted facilities.
- Access to facilities must be restricted for unauthorized personnel. Security personnel must be on site to provide monitoring for authorized access.

Roles & Responsibilities

Role	Responsibilities
Facility Management	● Responsible for designing and enforcing policies for facility control. ● Works across all departments and buildings to provide optimal working conditions for each facility, ensuring safety and security for all personnel ● Collaborates IT and security departments to provide authorized access to qualified and trained personnel, recording information on management system for personnel
IT	● Responsible for establishing system for authorized access for facilities that will require such system ● Ensures a robust security system that will permit authorized personnel to enter facilities, and deny entry to unauthorized persons
Security Department	● Responsible for enforcing security protocols ● Assist facility management in enforcing security for authorized access, and provide monitoring across all facilities for perimeter observation and entry/exit checks

Section XXVI: Management of Equipment

Introduction and Overview

The management of equipment at Cool Med Device Inc. is important because it helps to ensure that the equipment and applicable software associated can accurately and precisely test all medical devices that enter our facility. All equipment and software that is used in our facility that is used in data generation is required to be routinely monitored, calibrated, qualified, maintained, and controlled in order to ensure the equipment/software is working efficiently.

Scope

The purpose of this section is to outline the all qualification, calibration, maintenance, and control events of all software and equipment at Cool Med Device Inc.

Applicable SOPs

- SOP-QUAL-0437: Equipment Calibration and Maintenance
- SOP-QUAL-0420: Equipment Calibration Procedures
- SOP-QUAL-0421: Documentation Guidelines for Equipment Calibration
- SOP-QUAL-0435: Equipment Qualification Procedures
- SOP-QUAL-0436: Documentation Guidelines for Equipment Qualification
- SOP-QUAL-1899: Equipment and Software Master List
- SOP-QUAL-0430: Equipment Management
- SOP-QUAL-0431: Equipment and Software Installation

Policy

- All equipment and software will be managed, monitored, calibrated, and quantified by the system administration team through a master list that is located in Cool Med Device Inc's LIMS system.
- IT will assist System Administration for all software needs and ensure all software is secure and can properly generate data.
- All equipment and software will undergo the following qualification activities:
 - Installation Qualification: This is to ensure that the equipment and software is installed properly and can meet vendor specifications. This includes the correct equipment/software was delivered, confirm any components or accessories associated with the equipment are installed properly, equipment is installed properly, and documentation of the installation process along with any deviations.
 - Operational Qualification: This is to ensure that the equipment can operate and function properly and at specific parameters. The operational qualification will be used to verify operational parameters per vendor specifications to ensure proper performance. All results will be documented and any deviations from operational qualification will reflect if the instrument will be used for specific conditions only.
 - Performance Qualification: This is to help ensure that the equipment is working consistently and reliably in a quality control environment. The performance

qualification is performed using preparations as “mock samples” and will run performance qualification tests per vendor recommendations. All results will be analyzed to confirm the equipment can meet vendor specifications

- All qualification and maintenance items will be documented in real time and reflect the current status of the instrument with a Maintenance and Calibration Logbook to track all qualification and maintenance items.
- Should equipment or software be taken out of service and this should be documented in a Maintenance and Calibration Logbook and all troubleshooting actions should be documented in this logbook. All impacted user groups should be communicated in a timely manner on the instrument being taken out and returned to service.
- System Administration will perform monthly monitoring checks to ensure that all equipment is performing properly, check for signs of equipment wear and tear, record and review and data trends or issues, and ensure that all equipment and software is in compliance with operational standards.
- Routine maintenance will be performed on all software and equipment in order to help in the operational life of the software or instrument and prevent any potential equipment or software failures.
- Routine calibration of all equipment will be performed in order to ensure all equipment measures are reliable and accurate. All equipment calibrations will be performed using reference standards and will follow vendor specifications. Calibration results will be documented and equipment will be labeled with special calibration tags based on the results of the equipment calibration.

Roles and Responsibilities:

Role	Responsibilities
System Administrator	• Performs all qualification, calibration, and maintenance items for all equipment. • Performs routine monitoring of all equipment function and operation.
IT	• Performs and monitors all software systems to ensure safety, security, and accuracy of all equipment data. • Manages all records, service, incidents, and configurations of equipment software. • Restores and backup all software data as needed. • Perform periodic software system review. • Ensures all software can meet vendor specifications.

Section XXVII: Computer System Validation

Introduction & Overview:

Computer system validation (CSV) is a critical aspect of ensuring that all computer systems impacting GxP activities function as intended and comply with regulatory requirements. This section outlines the computer systems validated by Cool Med Device Inc, the process for conducting validations, the Validation Master Plan (VMP), and the roles and responsibilities involved in the validation process.

Scope:

This policy applies to all computer systems at Cool Med Device Inc. that support GxP activities, including systems used in manufacturing, quality control, data management, and regulatory compliance.

Applicable SOPs:

- SOP-QUAL-XXXX: Computer System Validation
- SOP-QUAL-XXXX: Validation Master Plan
- SOP-QUAL-XXXX: Change Control Procedure
- SOP-QUAL-XXXX: Risk Assessment Procedure

Policy:

- Computer Systems Subject to Validation:
 - Manufacturing Execution Systems (MES): Systems used to manage and monitor production processes.
 - Quality Management Systems (QMS): Systems used for document control, CAPA, audits, and nonconformance management.
 - Laboratory Information Management Systems (LIMS): Systems used to manage laboratory data and processes.
 - Enterprise Resource Planning (ERP): Systems used for managing business processes, including inventory and production planning.
 - Clinical Data Management Systems (CDMS): Systems used for managing clinical trial data.
- Validation Process:
 - Risk Assessment: Conduct a risk assessment to determine the extent of validation required based on the system's impact on product quality and regulatory compliance.
 - Validation Plan: Develop a detailed validation plan, including objectives, scope, responsibilities, and validation activities.
 - Installation Qualification (IQ): Verify that the system is installed according to specifications and that all necessary components are in place.
 - Operational Qualification (OQ): Test the system to ensure it operates correctly under defined conditions.

- Performance Qualification (PQ): Validate the system's performance under real-world conditions to ensure it meets predetermined criteria.
- Documentation: Maintain comprehensive documentation of all validation activities, including protocols, test scripts, and validation reports.
- Approval: Review and approve validation documentation by QA and other relevant stakeholders before the system goes live.
- Validation Master Plan (VMP):
 - The VMP outlines the overall strategy, objectives, and scope of computer system validation activities at Cool Med Device Inc.
 - The VMP includes a list of all systems subject to validation, the rationale for their validation, and a schedule for validation activities.
 - The VMP is reviewed and updated regularly to reflect changes in systems, processes, and regulatory requirements.

Roles and Responsibilities:

Role	Responsibilities
IT	● Develops and maintains the VMP. ● Coordinates and oversees validation activities. ● Ensures systems are validated according to the VMP and applicable SOPs.
Quality Assurance	● Reviews and approves validation plans, protocols, and reports. ● Conducts audits of validation activities to ensure compliance. ● Monitors the ongoing performance of validated systems.
System Administrator	● Executes validation activities according to the validation plan. ● Conducts IQ, OQ, and PQ testing. ● Maintains validation documentation.
Process Engineer	● Provides input on the validation requirements for new systems and changes to existing systems. ● Participates in risk assessments and validation activities.
Validation Team	● A cross-functional team including members from IT, QA, QC, and other relevant departments. ● Collaborates on validation activities, including planning, execution, and documentation.

Section XXVIII: Occupational Health and Safety Programs

Introduction & Overview

Facility management also encompasses the importance of occupational health and safety systems in the overall management process of facility control. The goal of these systems is to effectively integrate health and safety best practices (i.e. identifying hazards and controlling safety risks) into daily business workflow and operations, promoting a safe work environment for all personnel. This will ensure that Cool Med Device Inc.'s workplace is compliant with standards enforced by the Occupational Safety and Health Administration (OSHA).

Scope

This section details Cool Med Device Inc.'s implementation of occupational health and safety programs to adhere with OSHA's guidelines. Quality management defines occupational health and safety as a program to protect employees from work-related threats, hazards, and risks to safety. These programs will prioritize the maintenance of a safe, secure, and risk-free working facility for all personnel, visitors, contractors, and other employees.

Occupational health and safety protocols must adhere to standards enforced by ISO 45001, the standard that establishes several requirements for occupational health and safety management systems.

Applicable SOPs

- SOP-QUAL-0413: Environmental Monitoring Program
- SOP-QUAL-0424: Personal Protective Equipment (PPE)
- SOP-QUAL-0426: Sanitation and Sterilization Protocol
- SOP-QUAL-0427: Occupational Health And Safety System
- SOP-QUAL-0428: Procedure for Contaminated Products

Policy

- The primary goal of an occupational health and safety management system is to ensure that occupational illnesses, injuries, and diseases, and the aftereffects associated with them, can be prevented.
 - Prioritize employee health
- ISO 45001 specifies the requirements for a company's occupational health and safety (OH&S) management system and guides on implementation to company workflow. Cool Med Device Inc.'s OH&S management system should:
 - Fulfill legal requirements by OSHA
 - Promote productivity in completing OH&S tasks and performance
 - Achieve objectives within Cool Med Device Inc.'s OH&S department
- To ensure compliance across all personnel, OH&S department members must work with department heads to establish processes for consultations, team meetings, and participation of personnel to develop health and safety culture. This will allow the OH&S

management system to flourish and promote positive workplace culture as personnel are encouraged to maintain health & safety standards in the workplace.

- Hazard identification & prevention
 - Code of conduct and practice should be made clear and accessible to all personnel for handling of hazards detrimental to health & safety
 - Procedures for any hazard (i.e. biological, chemical, physical) must be documented and planned, and should be provided to personnel via training and departmental meetings
 - Continuous monitoring of how procedures are practiced and carried out is critical for identification of risks as soon as possible for best prevention
- Personnel must be informed of all policies surrounding OH&S, trained on hazard prevention practices, and made aware of standards enforced by ISO 45001 for the OH&S management system

Roles & Responsibilities

Role	Responsibilities
Occupational Health & Safety Department	● Responsible for designing and enforcing policies, protocols, and procedures encompassing occupational health & safety management system ● Work with all departments to develop and promote compliance plans for all personnel to contribute to health & safety culture
Facility Management & Quality Management	● Responsible for translation of OH&S policies to facility control ○ Ensures that facility maintenance protocols are aligned with main goal of providing safe and risk-free work environment for all facility inhabitants (i.e. personnel, visitors, contractors) ● Quality management will require documentation of protocols as SOPs for all personnel to refer to when training for hazard identification and risk management
Personnel	● Remain compliant with OH&S standards ● Learn hazard identification and risk management protocols during training periods, and collaborate with one another to promote healthy & safety culture throughout all working environments